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Next item →

1. Given the set $S = \{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$, which one of the following sets represents S using the set builder notation? 1 / 1 point

- ☐ $\{n^{-2} : n \in \mathbb{Z} \text{ and } 0 \leq n < 4\}$
- ☐ $\{2^n : n \in \mathbb{Z} \text{ and } 0 \leq n \leq 4\}$
- ☐ $\{\frac{1}{2^n} : n \in \mathbb{Z} \text{ and } 1 < n < 4\}$
- ☒ $\{2^{-n} : n \in \mathbb{Z} \text{ and } 0 \leq n < 5\}$

👉 Nice work

$$S = \{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\} = \{2^{-0}, 2^{-1}, 2^{-2}, 2^{-3}, 2^{-4}\}$$

2. Let S be a set. Which one of the following is always correct? 1 / 1 point

- ☐ $a \in S \text{ and } b \in S \implies \{a, b\} \subseteq \mathcal{P}(S)$
- ☐ $a \in S \implies a \in \mathcal{P}(S)$
- ☐ $a \in S \text{ and } b \in S \implies \{a, b\} \in S$
- ☒ $a \in S \text{ and } b \in S \implies \{a, b\} \subseteq S$

👉 Nice work

This satisfies the definition of the subset

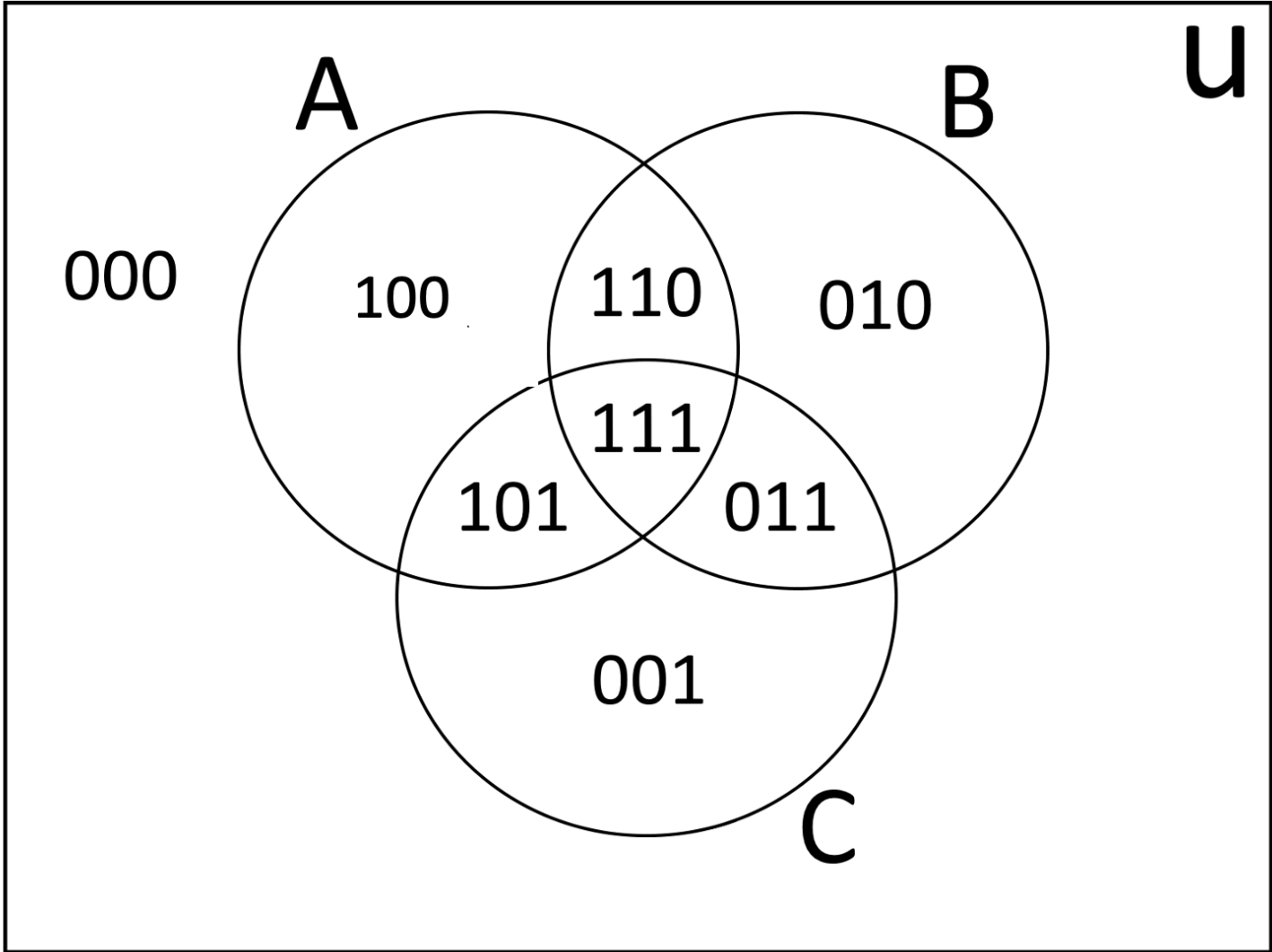
3. Let A and B be two sets. Which one of the following represents the set $(A - B) \cap B$? 1 / 1 point

- ☐ $A - B$
- ☐ $A \cap B$
- ☒ $\emptyset$
- ☐ A

👉 Nice work

$$(A - B) \cap B = (A \cap \overline{B}) \cap B = A \cap \overline{B} \cap B = \emptyset$$

4. Let A, B and C be three subsets of the universal set U . Given the following Venn diagram where each area is labelled by three binary digits, which one of the following represents the area 111? 1 / 1 point



(Image description: A Venn diagram with 3 circles: A, B and C. The area 000 contains not in the union of A, B and C. The area 100, contains the elements in A but not in the union of A and B. the area 010 contains the elements in B but not the union of A and C. The area 001 contains element in C but not in the union of A and B. The area 110 contains the element in the union of A and B but not in C. The area 101 contains the element in the union of A and C but not in B. The area 011 contains the elements in the union of B and C but not in A. The area 111 contains the element in the elements in the intersection of A, B and C. All three circles are set in a rectangle, labelled U.

- ☐ $(A \cup B) \oplus C$
- ☐ $(A \cup B) \cap C$
- ☐ $(A \cup B) - C$
- ☒ $(A \cap B) \cap C$

👉 Nice work

The area 111 is common to all three sets A, B and C

5. Applying De Morgan's laws, which of the following is equivalent to: 1 / 1 point

$$\neg(\forall x, P(x) \vee Q(x))$$

- ☐ $\forall x, \neg P(x) \vee \neg Q(x)$
- ☐ $\neg(\exists x, P(x) \wedge Q(x))$
- ☐ $\exists x, \neg P(x) \vee \neg Q(x)$
- ☒ $\exists x, \neg P(x) \wedge \neg Q(x)$

👉 Nice work

Excellent! You've correctly applied De Morgan's laws to both the quantifier and the logical operators.